

Session 4: PAKE-0, Assumptions, and Macros

Slides prepared by UMBC Protocol Analysis Lab

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Passwords

Password:

Log In

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Can help prove your identity to somebody.

We've used these forever.

Should be kept secret and hard to guess.

Often memorized. . .

Don't store these in plain-text anywhere.

PAKE-0

Alice

Bob

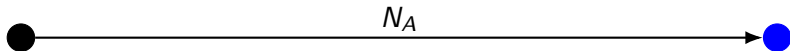
shared secret password: **pw**

● = Reception ● = Transmission

PAKE-0

Alice

Bob



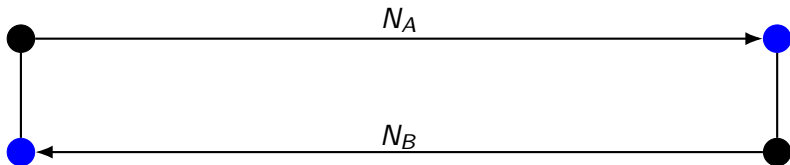
shared secret password: **pw**

● = Reception ● = Transmission

PAKE-0

Alice

Bob



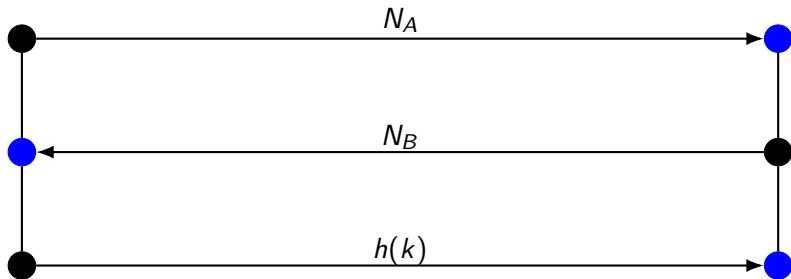
shared secret password: **pw**

● = Reception ● = Transmission

PAKE-0

Alice

Bob



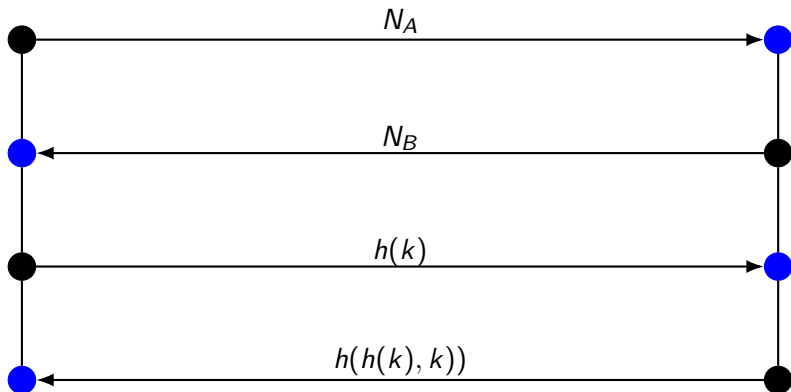
shared secret password: **pw** $k = h(pw, Alice, Bob, N_A, N_B)$

● = Reception ● = Transmission

PAKE-0

Alice

Bob



shared secret password: **pw** $k = h(pw, Alice, Bob, N_A, N_B)$

● = Reception ● = Transmission

PAKE-0 textbook diagram

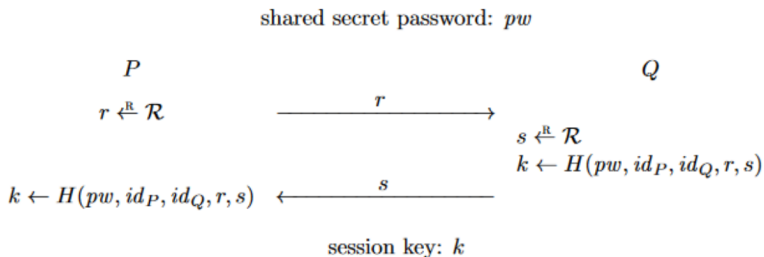
Figure 21.13: Protocol PAKE_0

Figure: Boneh, Dan, and Victor Shoup. “A graduate course in applied cryptography.” Draft 0.5 (2020).

Assumptions in CPSA (1)

Data origination assumptions:

uniq-orig (e.g., nonce)

Definition (unique origination)

A value *uniquely originates* in a bundle if it originates on exactly one strand. Note that if the unique origination point of a value is on a regular strand, then it cannot be produced in a “create” instance because that would constitute a second point of origination.

Assumptions in CPSA (2)

non-orig (e.g., private key)

Definition (non-origination)

A value is *non-originating* in a bundle if it does not originate on any strands. A simple lemma can show that a value is non-originating only if it is not carried in any message in the bundle.

Assumptions in CPSA (3)

pen-non-orig (e.g., pre-shared password)

Definition (penetrator non-originating)

A value is *penetrator non-originating* in a bundle of a protocol if it does not originate on any penetrator strands. Clearly, such a value cannot be produced in a “create” instance.

In-Class Activity: Model PAKE-0 in Small Groups

Now that you've seen the model, we will break you into groups with a TA assigned to each group to help and check your work. You will model this protocol using our new knowledge assumptions. instance.

Homework

Model more in CPSA

Model more in CPSA

Now that you've seen PAKE modeled, can you model a more challenging protocol?

Model your meeting protocol from Homework 1!

Survey

Due at 6:00pm on Tuesday the 28th

